Majorana zero modes in silicon: quantified failure of the conduction band
and a conditional hole-based route

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Abstract

We assess the feasibility of Majorana zero modes (MZMs) in silicon within a Bogoliubov–de Gennes framework whose validation chain, derivations, parameter tables, and convergence studies are given in the Supplementary Material. Conduction-band wires fail independently on three axes: spin–orbit coupling (best intrinsic topological gap $\sim 1.5\,\mu$eV; engineered SOC of $\alpha \gtrsim 0.02\,$eV Å required), a $g = 2$ ceiling forbidding induced gaps above $116\,\mu$eV at attainable fields (quantifying the optimal-coupling rule of Cole *et al.*), and valley physics—where we prove an exact two-band equivalence for smooth interfaces (Supplement S2) and derive that each single-atomic interface step acts as a near-π Josephson junction in the band-projected pairing ($\Delta\varphi = 2k_0a/4 \approx 0.85\pi$, Supplement S3), with a numerically determined bound state that caps the topological gap at $\sim$18% from the very first step. For the valence band we derive the gate-field covariation of $(\alpha, \hat{g}, m^*)$ from a four-band Luttinger–Kohn fin model: the field that builds direct-Rashba SOC (α up to $0.054\,$eV Å, spin–orbit lengths 32–101 nm, matching FinFET measurements) suppresses the wire-axis g below 1.1 while raising g_z to 2.4–4.2. Field-orientation maps then select the candidate designs. We label each input by its evidentiary status. *Measured*: Si:B critical fields ($B_{c2} \approx 0.1$–$0.4\,$T), FinFET hole l_{so} and g-tensors. *Derived*: the LK operating curve and all topological-gap numbers. *Hypothetical*: Pauli-limited thin-film Si:B and a hard-gap Al/Si-hole interface. On measured inputs only, a thick Si:B parent with tilted field supports 10–19 μeV. If 10–20 nm Si:B films reach their Pauli limit (a fabrication target consistent with measured coherence lengths, but not yet demonstrated), 21–27 μeV; if a hard-gap Al/Si-hole interface existed (none does), 34–55 μeV metallization-renormalized. The hole route is therefore a *conditional* feasibility result, decided by two standard measurements, not an established platform. Transport calculations show our spectral disorder thresholds are conservative, identify an abrupt class-D breakdown mode, and warn that step-disordered wires become Anderson insulators whose hard-looking transport gaps mask degraded local protection. Quasiparticle poisoning restricts the measured-field all-silicon stack to detection (not parity-coherent) experiments unless the parent gap is engineered. Two measurements decide the platform: the parallel critical field of thin Si:B films, and the wire-axis hole g-tensor in device geometry.

1 Introduction

Semiconductor–superconductor hybrid nanowires remain the most studied route to MZMs [1, 2, 3], and a decade of experiments—from the first InSb signatures [4] through the disorder-driven retractions and reappraisals [5] to protocol-based standards [6]—has made one lesson clear: material quality and honest accounting of trivial explanations dominate the problem. Silicon, the most controllable semiconductor in existence, has been conspicuously absent from this program, for reasons usually stated qualitatively: weak spin–orbit coupling (SOC), a small g-factor, and conduction-band valley degeneracy.

This paper makes those reasons quantitative, finds that one of them hides a qualitatively new failure mechanism, and shows that none of them applies to silicon *holes*. Our contributions: (i) a quantified electron verdict (Secs. 3.1, 3.2); (ii) an exact unitary equivalence making smooth-interface valley splitting benign under the physical inter-valley pairing channel, and the identification of single-atomic interface steps as near-π Josephson junctions that destroy protection from the first step (Sec. 3.3); (iii) a Luttinger–Kohn (LK) derivation of the gate-field covariation of hole parameters, which erodes free-parameter optimism but preserves a viable design (Sec. 4); and (iv) orientation- resolved feasibility maps for two silicon-compatible parents—laser-doped superconducting Si:B [18, 19, 21, 22] and Al films—including thin-film design targets, metallization renormalization, and quasiparticle-poisoning limits (Sec. 4.2). Throughout, spectral diagnostics are cross-checked by recursive-Green's-function transport (Sec. 5).

2 Model and methods

We use the standard single-channel BdG Hamiltonian

$$H = \left(\frac{\hbar^2 k^2}{2m^*} - \mu + V(x)\right)\sigma_0 + \alpha k \sigma_z + E_Z \sigma_x \quad \text{(normal part)}, \qquad \Delta\, i\sigma_y \quad \text{(pairing)}, \tag{1}$$

discretized with exact particle–hole symmetry by construction, topological for $E_Z^2 > \Delta^2 + \mu^2$. Validation (repository Fig. 1): gap closing at the analytic boundary; end-localized zero modes; exponential length splitting; finite-wire vs analytic bulk-gap agreement. Valley physics uses an $8N$ basis with *inter-valley* singlet pairing $\Delta(\nu_x \otimes i\sigma_y)$—the physical zero-momentum channel since the relevant valleys are time-reversed partners—with arbitrary complex valley-orbit (VO) profiles $\lambda(x)$ and a gauge-covariant phase-locked Dresselhaus-type SOC option. Hole parameters derive from a four-band LK model of the fin cross-section (validated against exact [100] bulk masses) with vertical field, bare (κ) and orbital Zeeman terms. Transport uses a dimension-generic RGF with Ando wave-matched normal leads and an $E = 0$ reflection invariant $\operatorname{sign}\det r$ (kwant-equivalent; validated against closed-chain spectra to 10^{-13}). A test suite pins every exact claim. Δ is a *static* induced gap (the $\omega = 0$ limit of the standard tunneling self-energy); parent-gap field suppression enters via $\Delta_{ind} \le \Delta_p(B)$ and metallization via the quasiparticle weight Z (Supplement S6) — we do not solve the dynamical self-energy problem, and the renormalized numbers bound the direction of the omitted physics. All caricature choices (GL/Zeeman-linear parent gaps, iid onsite disorder with correlation length dx, hard-wall LK geometry) are stated where used; complete per-figure parameter tables, seed families, and discretization/length/seed/grid convergence studies are in Supplement S4–S5.

3 Conduction band: three independent failures

3.1 Spin–orbit shortfall

Maximizing the topological gap over ($\mu, B \le 2\,\mathrm{T}$) at measured intrinsic Rashba strengths gives $0.16\,\mu\mathrm{eV}$ (typical, $\alpha = 10^{-4}\,\mathrm{eV\,\AA}$) to $1.5\,\mu\mathrm{eV}$ (optimistic, 10^{-3}): below dilution-fridge protection scales and destroyed by any realistic disorder. Reaching a robust $20\,\mu\mathrm{eV}$ requires $\alpha \approx 0.015$ (idealized) to $0.021\,\mathrm{eV\,\AA}$ (with parent-gap suppression $\Delta(B)$)—an optimistic lower bound given unmodeled orbital and metallization corrections. Micromagnet arrays [8, 9] reach this scale in principle; a 2025 InAs experiment measured synthetic $\alpha = 0.022\,\mathrm{eV\,nm} = 0.22\,\mathrm{eV\,\AA}$ [10]—an order of magnitude *above* our silicon requirement, though demonstrated in a lighter-mass material with a mature hybrid stack, and with micromagnet fringe-field disorder unmodeled here. "Engineered-electron silicon" is therefore marginal rather than absurd—but the valley physics below stacks on top of it.

3.2 The $g = 2$ catch-22

Since $E_Z(2\,\mathrm{T}) = 116\,\mu\mathrm{eV}$ for $g = 2$, any induced gap $\Delta \geq 116\,\mu\mathrm{eV}$ can never reach the topological phase at attainable fields; in the SOC-limited regime relevant to engineered devices, increasing interface transparency *reduces* the achievable gap, with an optimal parent gap $\Delta_0 \approx 98\,\mu\mathrm{eV}$ once $\Delta(B)$ suppression is included. This sharpens, for $g = 2$, the weak-coupling optimum identified by Cole, Das Sarma and Stanescu [7]. Pauli-limited parents satisfy the constraint automatically (Sec. 4.2).

3.3 Valley physics: an exact equivalence and step junctions

Equivalence theorem. With inter-valley pairing and *any* uniform VO phase, the wire maps unitarily onto two decoupled bands at $\mu \mp |\lambda|$ with pairing $\pm\Delta$ (proof in Supplement S2; the key step is the exact invariance of the inter-valley pairing under the valley rotation, $U_\phi D U_\phi^T = D$; verified numerically to $3 \times 10^{-10}\,\mu\mathrm{eV}$ for valley-scalar and phase-locked Dresselhaus SOC alike): smooth interfaces are benign, and valley splitting $\gtrsim 100\,\mu\mathrm{eV}$ buys single-valley topology, as the simple wedge picture suggests.

Step junctions. Real interfaces are stepped. That steps shift the VO phase by $2k_0\delta z$ is established quantum-dot physics [12, 11]; the new statement here is its consequence under inter-valley *pairing*: projecting the pairing onto the local valley bands gives each band an order parameter of phase $-\phi(x)$ (Supplement S3, exact), so each single-atomic step, $\Delta\varphi = 2k_0(a/4) \approx 0.85\pi$, is a Josephson junction accidentally close to π in silicon. The bound-state depth is determined numerically; the transparent short-junction formula $\Delta_t|\cos(\Delta\varphi/2)|$ tracks it to 10–25% (Supplement Table 1). A controlled comparison (same-sign staircase vs sign-randomized steps of equal magnitude) shows net phase *winding* is irrelevant at realistic densities; the damage is per-step junction physics: a single step binds a subgap state at $4.0\,\mu\mathrm{eV}$ (18% of the $22\,\mu\mathrm{eV}$ clean gap), deepening to a Kitaev domain-wall zero mode at $\Delta\varphi = \pi$ [Fig. 1(b)]. Ensembles at 50 nm step spacing retain only $\sim$0.7–2.3 $\mu\mathrm{eV}$ (median, L-dependent upper bounds). Bunched double-height steps ($\Delta\varphi = 1.7\pi \equiv -0.3\pi$) are $\sim 2\times$ milder—step-bunched vicinal surfaces are preferable—while VO-amplitude suppression near steps [12] worsens the damage; wires laid along step edges avoid it entirely ($|\cos\chi|$ scaling). Smooth phase gradients are a real but subdominant second channel, suppressed at large valley splitting. Transport (Sec. 5) adds a warning: step-dense wires become Anderson insulators whose end-to-end transport gap *grows* with length ($0.5/5.6/23\,\mu\mathrm{eV}$ at $L = 1.5/3/6\,\mu\mathrm{m}$) while localized subgap states persist—a hard-looking transport gap does not certify protection. The falsifiable consequence: induced-gap hardness in proximitized Si *electron* devices should degrade with miscut/step density from the very first step, testable by gap spectroscopy on standard miscut wafer series; holes are immune by construction.

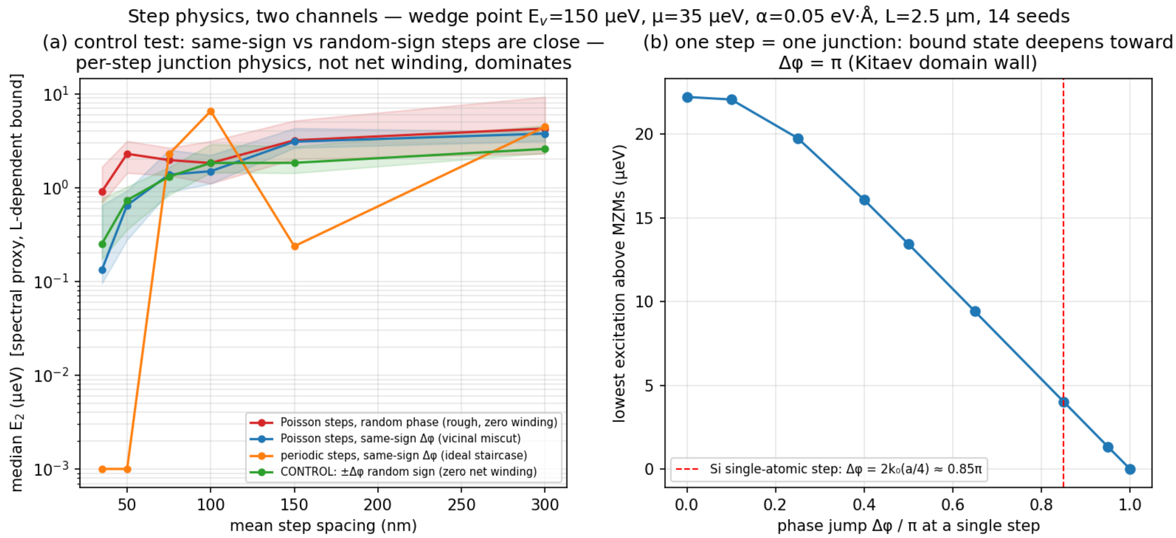

Figure 1: Step physics in the inter-valley-paired electron wire. (a) Controlled scenario comparison at fixed jump magnitude: same-sign (vicinal) and sign-randomized (zero-winding control) staircases are statistically indistinguishable—per-step junction physics, not winding, dominates. (b) A single VO phase step is a Josephson-like junction; the Si single-step value 0.85π sits near the Kitaev domain-wall point π.

4 Valence band

4.1 Luttinger–Kohn constraints: the α–g covariation

Quasi-1D hole channels in Si FinFETs have measured strong, gate-tunable SOC [15, 17, 14] and anisotropic g-tensors [16]. Our four-band LK fin model reproduces the measured spin–orbit-length window (l_{so} = 32–101 nm) and yields the constraint the free parameter box hides [Fig. 2]: the vertical field that builds α (up to 0.054 eV Å) suppresses the wire-axis g_x from 1.1 to 0.5 while raising g_z to 2.4–4.2, with $m^* \approx 0.44$ and the SOC axis in-plane transverse. A seven-geometry bracket (7–16 nm cross sections, 3–30 MV/m) keeps $g_x \in [0.4, 1.1]$: the measured $g_{xx} \approx 1.9$–2.3 [16] is *not* reproducible in four-band hard-wall LK, so all platform numbers below are carried under both the LK and empirical tensors; resolving the discrepancy requires six-band $k \cdot p$ with device electrostatics (top remaining theory item).

4.2 Parents, orientation, and the surviving designs

Field-orientation maps [Fig. 3] under both g-tensors and two parents give the design rules: (i) an **Al film** parent is locked to in-plane fields but strong—34–55 μeV metallization-renormalized—*if* a hard-gap Al/Si-hole interface can be fabricated (none exists); (ii) a **thick Si:B** parent [18, 19, 20, 21] is orientation-tolerant and, using only measured critical fields ($B_{c2} \approx 0.1$–0.4 T, orbital-limited), supports 10–19 μeV at fields tilted toward out-of-plane (riding g_z; the center point is 11.0 μeV bare / 10.1 renormalized on the finest optimizer grids, Supplement S5, and moves over 8–11 μeV under $\pm$20% parent-model variations); the in-plane wire-axis direction is g-starved (3–9 μeV). (iii) The **thin-film Si:B** route turns the "Pauli-limited" hypothesis into a fabrication target: with coherence lengths from measured perpendicular critical fields (ξ = 29–57 nm), films of $d \lesssim$ 18–36 nm have parallel orbital critical fields above the Pauli limit B_P = 1.11 T, and d = 10–20 nm films support

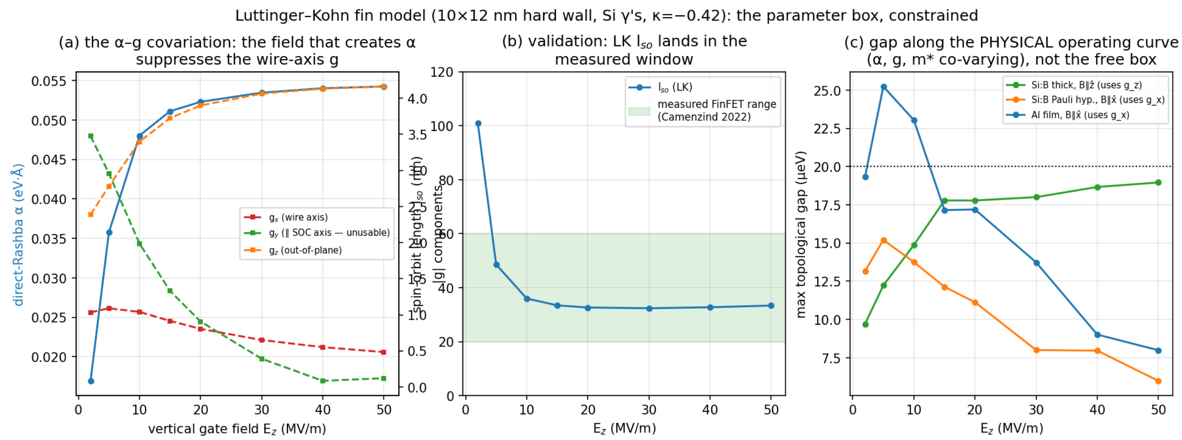

Figure 2: Luttinger–Kohn constraint on the hole platform: (a) α–g covariation under the gate field; (b) validation of l_{so} against FinFET measurements; (c) accessible gap along the physical operating curve for three parent/orientation scenarios.

21–27 μeV (modulo unmodeled $T_c(d)$ suppression). Quasiparticle poisoning bounds the role of the measured-field stack: at operating fields the Si:B parent gap is 29–33 μeV, so $x_{qp} = 10^{-6}$ requires $T_{\text{eff}} \leq$ 25–29 mK (Al reference: 130 mK)—a detection platform, not a parity-coherent qubit, absent parent-gap engineering.

5 Transport-level verification

RGF transmission with normal leads validates the clean wire (transport gap 37.3 vs bulk 36.8 μeV) and provides sign det r as a topological invariant. Under onsite disorder the spectral proxy (third-smallest $|E|$) is *conservative* at moderate strength (localized states carry no current): the engineered-electron case's half-gap threshold moves from $W \approx 410$ to $\approx 548\,\mu$eV, and breakdown is abrupt—a class-D zero-energy transmission anomaly coinciding with invariant sign flips—rather than a soft closing [Fig. 5]. The same machinery extended to eight-orbital valley cells yields the Anderson-insulator warning of Sec. 3.3.

6 Discussion

Two measurements now decide the silicon question. First, the **parallel critical field of thin (≤20 nm) laser-doped Si:B films**: if it approaches the Pauli limit, as measured coherence lengths permit, the all-silicon hole stack supports $\gtrsim$20 μeV gaps in a CMOS-compatible, same-crystal materials system; if films remain orbital-limited at $\sim$0.4 T, the stack is capped near 10–15 μeV and detection-only. Second, the **wire-axis hole g-factor and SOC axis** in the actual gated geometry: our LK bracket and the published FinFET values differ by 2× along the one direction that matters for film parents. Both measurements are standard characterization, neither requires a working topological device, and the electron-side miscut prediction offers a third, independent test of the framework. We caution that the step-junction mechanism, the orientation maps, and all platform numbers inherit the stated caricatures (single channel, hard-wall LK, parent-gap models, iid disorder); their domains of validity and the sensitivity of every headline number to them are quantified in Supplement S5–S6.

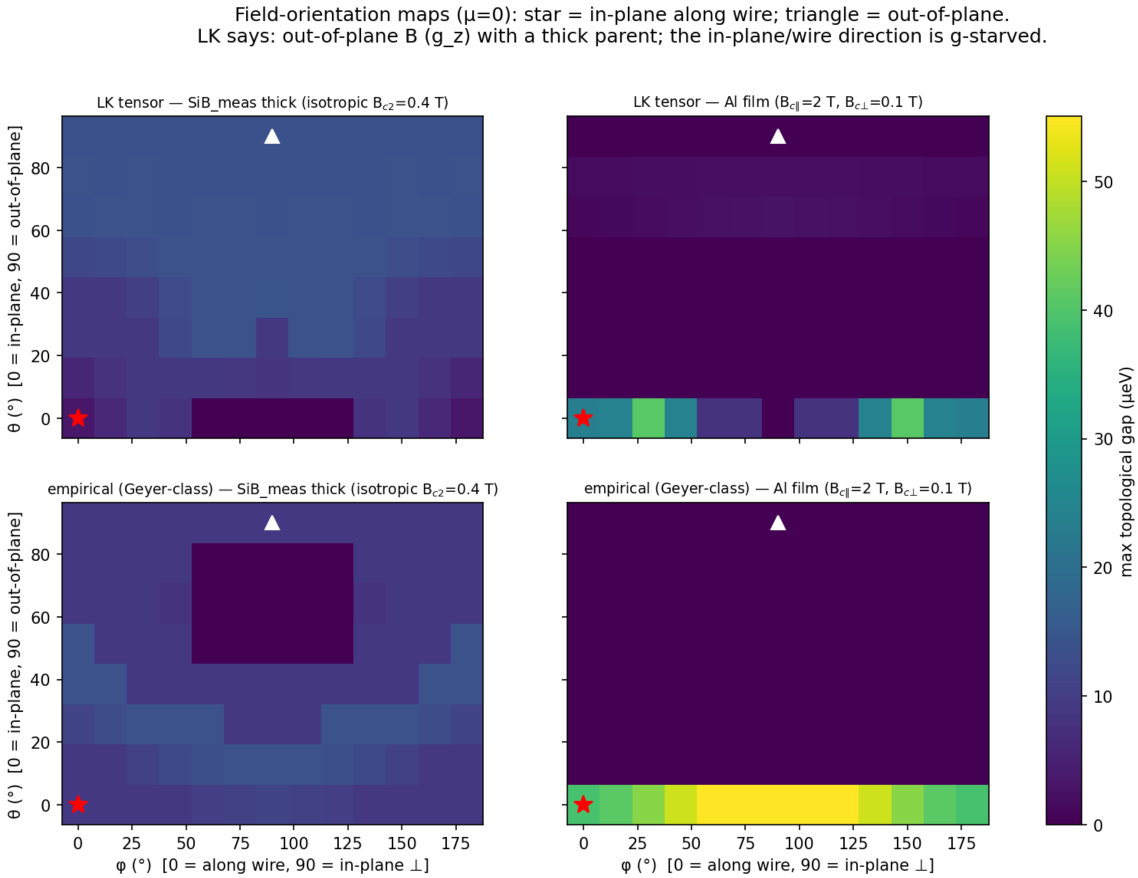

Figure 3: Maximum topological gap vs field orientation for LK (top) and empirical (bottom) hole g-tensors, with thick measured-Si:B (left) and Al-film (right) parents. Star: in-plane along the wire; triangle: out-of-plane.

Data and code availability

All figures and quoted numbers regenerate from the committed code (`run_analysis.py`, `transport.py`, `transport_valley.py`, `lk_holes.py`, `qp_poisoning.py`, `convergence.py`, `tests.py`); the numerical ledger is `output/data/key_numbers.json`, and per-figure regeneration commands, the environment specification, and seed families are documented in `README.md` and Supplement S4/S8. An archival snapshot (repository URL, commit hash, and DOI) will be deposited on Zenodo at submission. [Zenodo DOI: to be inserted at submission.]

Acknowledgments and provenance

Analysis, numerical implementation, verification, and manuscript preparation were assisted by AI agents (Claude, Anthropic) under the author's direction. Three internal numerical-review rounds and one round of revisions in response to external review are documented, with their corrections, in the repository log. AI-assisted review supplements but does not substitute for the reproducibility materials above; all claims trace to committed, seeded code.

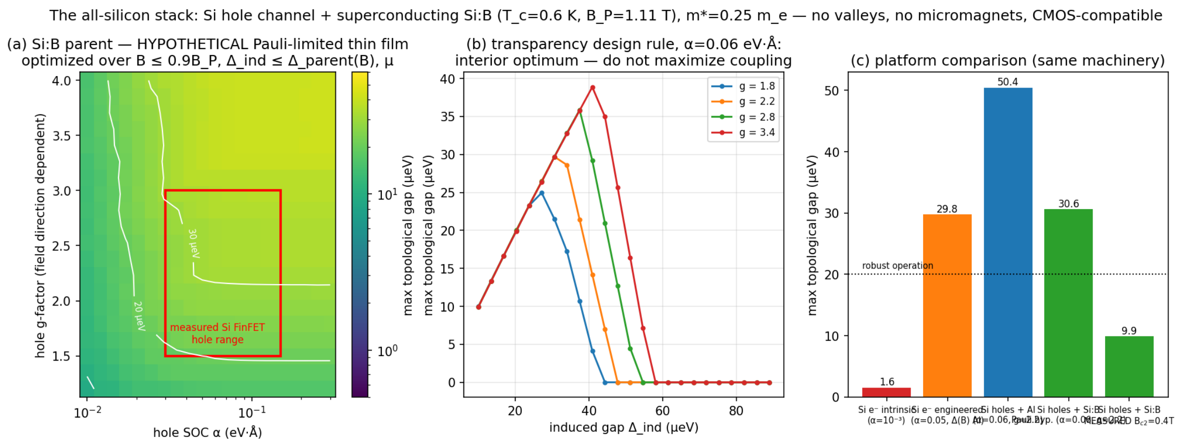

Figure 4: The all-silicon stack under the (hypothetical) Pauli-limited Si:B parent: (a) gap over the empirical (α, g) box; (b) the transparency design rule (interior optimum); (c) platform comparison including the measured-field Si:B scenario.

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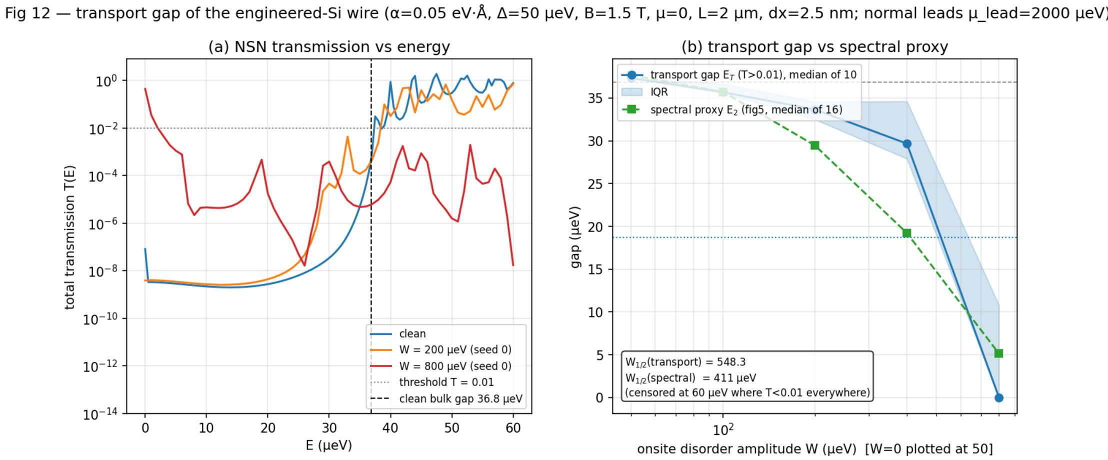

Figure 5: Transport vs spectral disorder thresholds (engineered-electron case): (a) transmission spectra; (b) median transport gap vs disorder with the spectral proxy overlaid.

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